GENERAL THEORY AND GOOD PRACTICES IN ECOLOGICAL NICHE MODELING: A BASIC GUIDE

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Introduction

Ecological niche modeling (ENM) has become a tool of intense attention in the past two decades (Guisan et al. 2013, Peterson et al. 2011, Peterson et al. 2018) with a concurrent development of the breadth of questions to which it is applied, ranging from species range delimitation, studies of the effects of climate change on species suitable areas, to the study of niche evolution (Luna-Aranguré and Vázquez-Domínguez 2019, Peterson et al. 2011, Warren 2012). Niche modeling attempts to establish a relationship between occurrence records and environmental variables in order to predict the suitability of habitat of a studied species (Peterson et al. 2011, Warren 2012). For this reason, ENM studies also have been applied to answer questions related to different aspects of biogeography, to predict the existence of unknown populations and species, to understand the processes of invasion, disease risk explorations, among others (Peterson 2006, Peterson 2014, Yates et al. 2018).

Of particular importance for ENM and other similar implementations in species distribution modeling (SDM), is the theoretical framework that considers biotic interactions (**B)**, environmental conditions (**A**), and mobility (**M**) for understanding species distributions (the BAM diagram; Soberón and Peterson 2005). Using this framework, the actual distribution of a species represents sites where three conditions exist: positive biotic interactions, suitable environmental conditions, and accessible areas. That is, areas in geography (**G**) corresponding to **B** ∩ **A** ∩ **M**. These considerations are of critical importance when performing modeling exercises; for example, they help to determine the areas to be considered when calibrating ENMs (areas inside **M**). More detailed descriptions of the importance of the BAM diagram, as well as, the types of ecological niches and distributional areas that can be identified using this framework already exist (see Peterson et al. 2011, Saupe et al. 2012). A more detailed description of the area **M** is given in Section 1.2 since its definition is crucial to perform some steps in ENM/SDM exercises.

The steps and materials required for the construction of an ENM rely heavily on the purpose of the study. Nevertheless, four essential items remain constant in any ENM study: (1) occurrence data, (2) geographic region of study, (3) environmental variables, and (4) an algorithm to generate the model. Here we describe how to integrate those four items into the modeling process through the following steps (Fig. 1): dataset preparation (Section 1), model construction (Section 2.1), and post-modeling processing (Section 2.2–2.4). For the environmental dataset preparation, we will provide a guide with different approaches on how to select the best set of variables to calibrate a model. Then, we move on how to define and perform model calibration. Under model construction, we will describe the best practices to select different features for model calibration using Maxent (e.g., regularization multipliers, feature classes, etc.). Finally, we will describe the processing of the models created, and how to assess and interpret threshold values to define suitability limits. Note that we will not cover model extrapolation in this manual at its full extent. In addition, steps dealing with occurrences and data cleaning (Fig. 1) were already addressed in a previous work (Cobos et al. 2018).

Due to the popularity and friendly interface, we chose to illustrate and perform most of the steps cited here using QGIS[1] as geographical information system (GIS) and the open-source modeling software Maxent to perform the ENM/SDM analysis itself (Phillips et al. 2006, Phillips et al. 2017[2]). In Section 1, we included a detailed description of each kind of data needed before starting the analysis. All these pieces of information are usually stored in different format files, including "csv", "ascii" raster grid (format "asc"), and shapefiles (format "shp"); thus, different tools are needed to clean, visualize, and analyze the datasets. Given the type of data and software used in this manual a basic knowledge of how to use GIS tools is required.

[1] http://www.qgis.org.

[2] https://biodiversityinformatics.amnh.org/open_source/maxent/.

We are aware that there is a plethora of algorithms available to estimate species distributions (e.g., BIOCLIM, random forest, general linear models, etc; Qiao et al. 2015). However, for the sake of simplicity and practicality, we demonstrate the analysis here in Maxent, as our objective is merely to provide practical guidance on the basic steps of developing ENM/SDM. The distributional and environmental dataset used to perform the steps of this manual are included in the same download.

It is worth mentioning that the procedures described can also be implemented with the programming language R, as a wide variety of packages are available to perform the modelling steps (e.g., *dismo*, *ggmap*, *ggplot2*, *kuenm*, *raster*, etc.). We acknowledge that programming skills might be perceived as a challenging effort for some users, which will narrow the potential audience that benefits from the present guide. Thus, we leave the users free to use other resources and determine the appropriate methods for their research (see Supporting information S1 for a list of alternative software that can be used to perform ENM/SDM analysis). For those familiar with the R language, we encourage them to perform the different steps presented in this manual using the scripts provided in the Supporting information S2. We also would like to mention that this guide presents step-by-step indications of most procedures; however, these procedures can be done using distinct tools and in different ways.

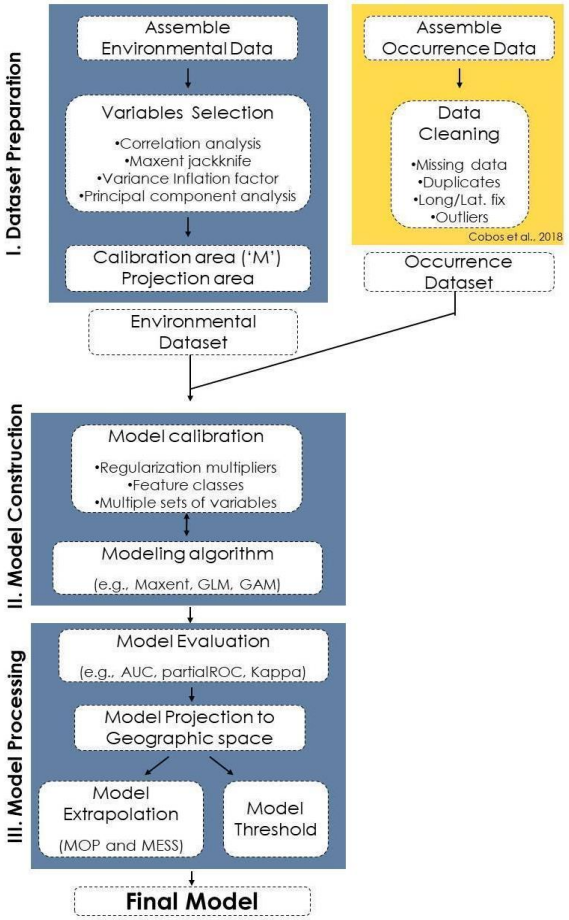


Figure 1. Summary of steps in the overall species distribution modeling process. Scheme adapted from Peterson et al. 2011. See text for detailed description.

Note: To refer to this manual please use,
Simões, M., Romero-Alvarez, D., Nuñez-Penichet, C., Jiménez, L., Cobos, M.E. 2020. General theory and good practices in ecological niche modeling: a basic guide. http://hdl.handle.net/1808/30276

1. Datasets required to perform an ENM/SDM analysis

In this section, we describe the first steps to build an ENM/SDM: data assembly, cleaning, and formatting. Two types of data are required: (i) occurrence data, documenting presence (and sometimes absence, for other algorithms) of the species, and (ii) environmental variables that define the scenopoetic *n*-dimensional niche (Hutchinson 1957) that describe the ecological requirements of the species. In section 1.1, we describe the occurrence data format used to fit the model. In section 1.2, we describe how to build the calibration area to create the model (**M**; Barve et al. 2011, Soberón and Peterson 2005), and in section 1.3, we describe different methods for selecting the predictor variables. A practical example describing each part of the process is presented at the end of each section.

1.1 Occurrence data

ENM applications require information regarding the distribution of the species (occurrence records) and environmental data that characterize the landscape across which the species is distributed. One could argue that the exponential growth of ENM/SDM applications in the past 20 years (Guisan et al. 2013) is due to the increasing availability of primary biodiversity data—principally in the form of specimen information—owing to advances in information technology and in the policies of owners of primary data sources (Soberón and Peterson 2004).

The Global Biodiversity Information Facility (GBIF; http://www.gbif.org) is the largest online provider of distribution records, with free access to digitized data from different sources (e.g. museum collections, survey programs, etc.). At present, GBIF provides rapid access to more than a billion occurrence records, allowing fast access in comparison with compiling data from original sources, and making large-scale taxa analysis feasible in relatively short time frames (Beck et al. 2012, Jetz et al. 2012); although its content has been strongly criticized due to data quality issues (Graham et al. 2004, Yesson et al. 2007). Therefore, these data should be used with caution and preliminary processing before any further use.

The organism used to exemplify ENM/SDM steps in this manual is *Dendrobates auratus* (Girard, 1855), also known as the green-and-black poison dart frog. This species is found in the humid lowlands from southeastern Nicaragua to northwestern Colombia (IUCN 2019). The dataset used to characterize its distributional range was extracted from the GBIF database (2598 occurrence records) and it went through a series of steps of data cleaning and error detection (for step-by-step manual on cleaning biodiversity datasets see Cobos et al. 2018). The final dataset used here contains 188 occurrence records (**D_auratus.csv**) covering a region from Costa Rica to northern Colombia (Fig. 2). The file contains the presence localities; its first column contains the species name and the following two columns contain the geographic coordinates of each occurrence record, mandatorily, in various tools (e.g., raster package), organized in the order: “longitude” and “latitude” (Fig. 2, red arrows).

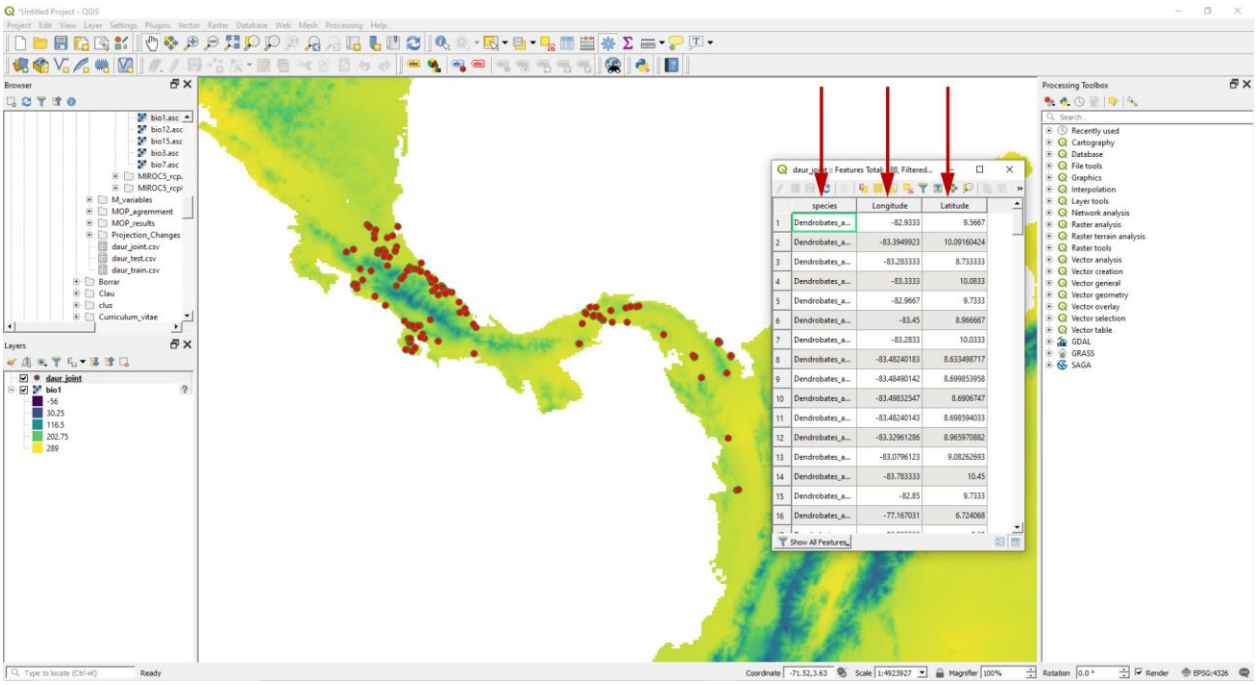


Figure 2. QGIS window showing occurrence records for *Dendrobates auratus* (blue points) represented on top of a raster of mean annual temperature. On the right panel is the attribute table for "D_auratus.csv", showing the species name, longitude, and latitude (red arrows). Note that the species name is separated by an underline. It is advisable to avoid empty spaces or special characters (e.g., accents) in text files such as "csv" files, as some software recognize them as corrupt characters.

1.2 Calibration area (**M**)

Once the occurrence data has been cleaned, the next step is to define the calibration area for the studied species, also known as the **M** region. According to the **BAM** diagram (Soberón and Peterson 2005), the **M** represents areas that have been accessible to the species during its biogeographic history. This area must be carefully delimited by considering the biology and dispersal abilities of the species, as well as, the environmental conditions represented in the area. The justification for the delimitation of the **M** region should be done explicitly, as it will impact all aspects of the analysis, including model parameterization, validation, and comparisons (Anderson and Raza 2010, Barve et al. 2011). For instance, large calibration areas tend to inflate evaluation metrics and to bias conclusions (Barve et al. 2011, Lobo et al. 2008, Peterson et al. 2008).

As a straightforward example, when modeling suitable areas or potential distributions of species with known restricted range, such as the blue-legged mantella (*Mantella expectata*), native from South Madagascar, we may choose to define the calibration area as the whole island. This area (the island) includes the home range of the species, and areas that populations of this frog have been able to occupy during the species' life history. We will make sure to not include regions outside Madagascar, as they have never been accessible to the species.

On the other hand, the delimitation of **M** for a poorly known or invasive species remains controversial (Owens et al. 2013), and the question on how to define the calibration area is still unresolved. Nevertheless, we encourage to describe explicitly the way a calibration area has been defined when presenting a study (Barve et al. 2011, Owens et al. 2013, Peterson et al. 2011, Peterson 2014, Saupe et al. 2012) to assure replicability and transparency during the ecological niche modeling process.

1.2.1. Creating the **M**

A hypothesis for delimiting **M** can be derived using different proxies of what we know about the species. Valid approaches include: polygons delimiting all the peripheral occurrence points (i.e., convex polygons), polygons representing biogeographic and environmental features (e.g., ecoregions), buffers surrounding all the occurrence points, among others. When buffers are applied to occurrences or polygons such as the ones described above, buffer distances could be based on measurements such as the largest distance between occurrences (e.g., Romero-Alvarez et al. 2017), the mean distance of all occurrences to their centroid (e.g., Poo-Muñoz et al. 2014), or a distance based on the species' dispersal ability, for instance. Further considerations could lead to delimit the areas previously mentioned only to regions with enough sampling effort (Peterson 2014). In this context, using the boundaries of a country where the studied species has been thoroughly investigated as a calibration area, could be a valid approach (e.g., Gurgel-Gonçalves et al. 2012), although is not recommended as the only criterion to define the **M**.

GIS software is commonly used to define the **M**. QGIS is one of the tools that allows manual creation of an area as a polygon shapefile ("shp"), which will be used to crop and mask the environmental layers. Here, we go through the steps to create a calibration area by buffering the occurrences to create a shapefile that will be then used to mask the variables for calibrating the model of *D. auratus*:

1. In QGIS, convert the "csv" file containing the distribution of *D. auratus* into a shapefile, by right clicking on the occurrences "Export > Save features as". Export the shapefile to the same directory you are working and import it to QGIS (most of the time it will be imported directly).
2. In QGIS, under "Vector geometry tools", go to "Buffer" (Fig. 3). Select the shapefile you created under the field "Input layer", select the output folder under "Buffered", and select the name (Fig. 4). In our case, we will use a 50 km (~0.45 decimal degrees) buffer around each occurrence data due to the limited dispersal capacity of the species (Whitfield 1984). As previously mentioned, the delimitation of the calibration area is a relevant step during the modeling process. Moreover, it is a hypothesis and should be explicitly explained.

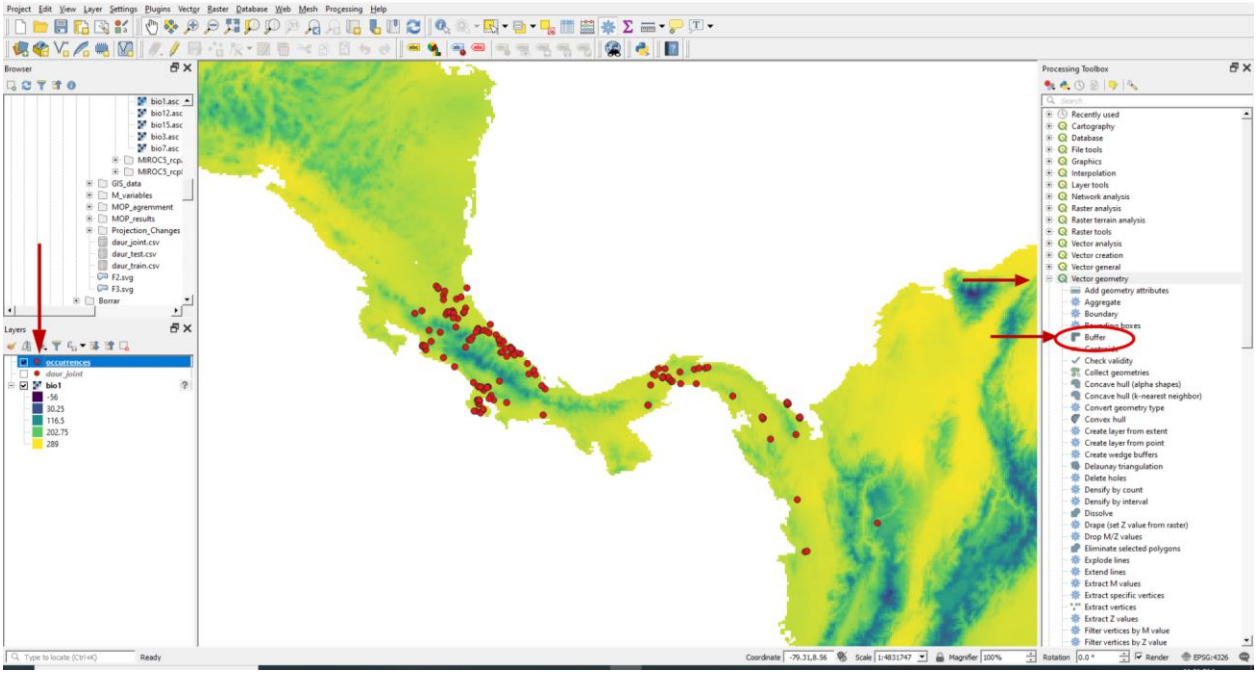


Figure 3. QGIS window showing the Toolbox panel with the Buffer tool. Red circle indicates the tool used to convert the polygon to shapefile.

3. Along with creation of buffers for each occurrence point, the polygons need to be dissolved, so we can obtain a single area that will characterize the **M** of the species. For that, activate the option “Dissolve result” (Fig. 4) which will create a new file with a single region (Fig. 5).

R scripts to perform the creation of **M** (steps 1–3) can be found in the links provided in the Supplementary Material S2.

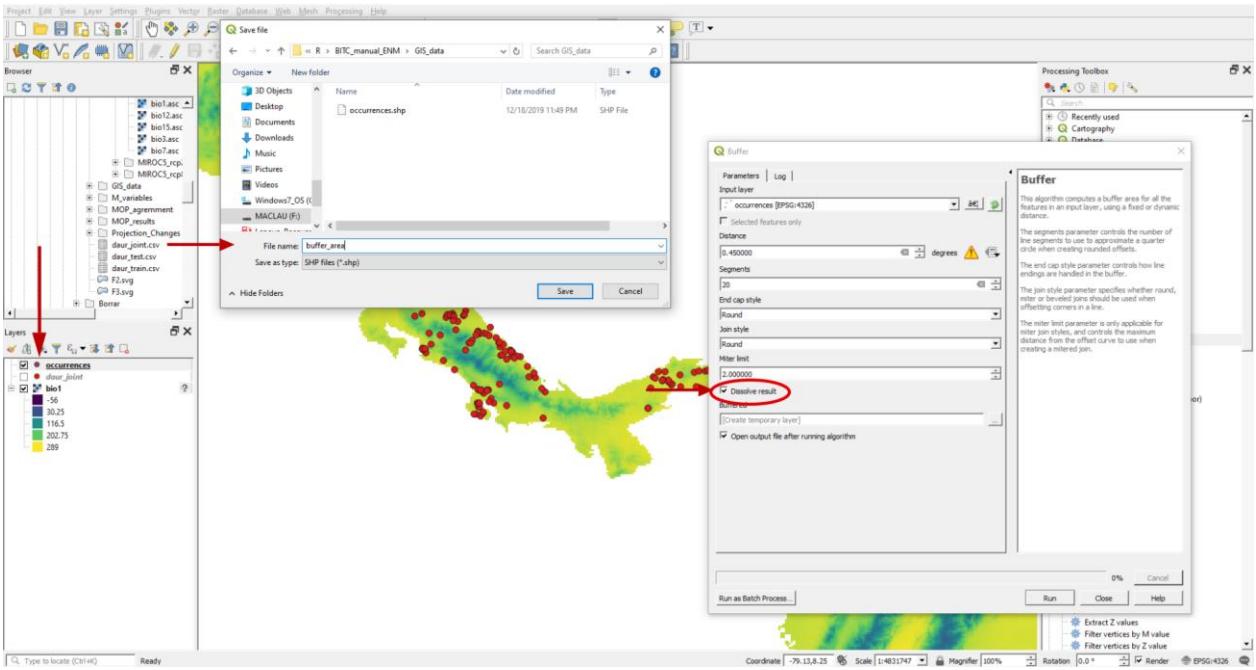


Figure 4. QGIS window showing how to create a file with buffers of 50 km (~0.45 decimal degrees) around each occurrence data point. Dissolving is done using an extra functionality of the same tool (red circle).

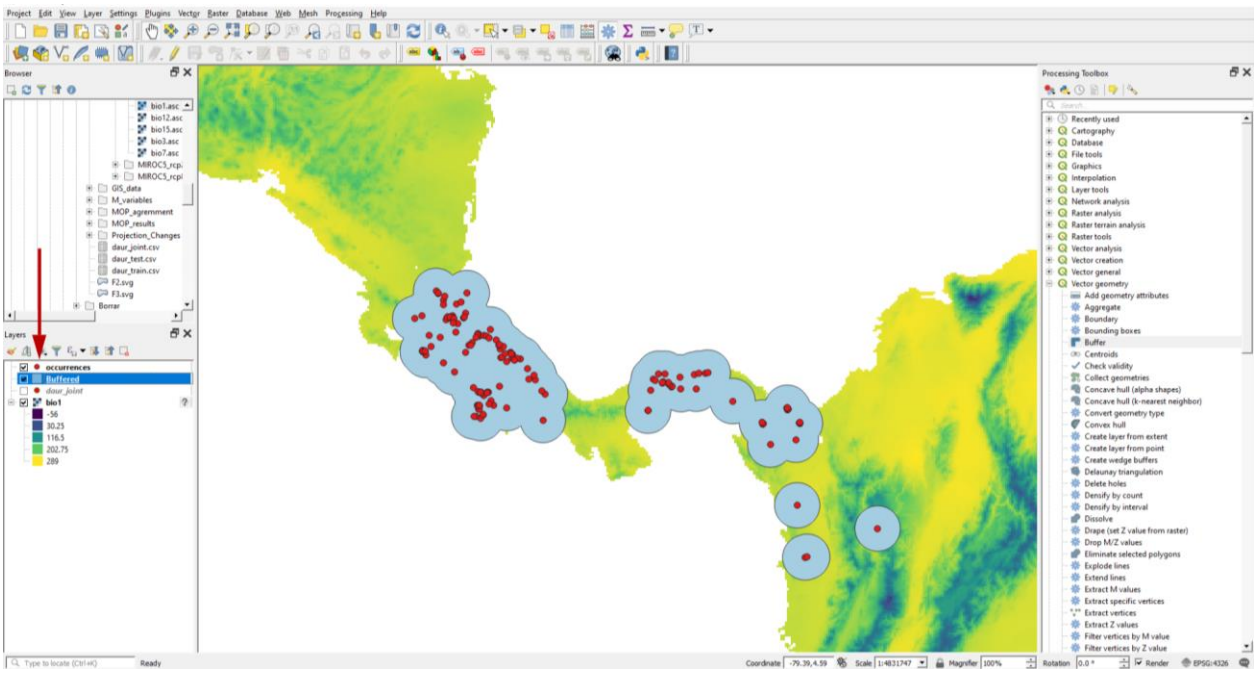


Figure 5. QGIS window showing the dissolved buffers that will be used to characterize the calibration area of *Dendrobates auratus.*

1.3 Predictors/Environmental layers

Predictors represent the different dimensions used to characterize species niches in a multivariate environmental space. The ability to model a species' geographic distribution, given occurrence records and environmental information, assumes that abiotic factors directly or indirectly affect species distributions (Austin 2002). These abiotic variables are represented by **A**, in the BAM. Environmental data (e.g., temperature, precipitation, etc.) have been the main predictors in ENM/SDMs (for examples of other types of variables see also Freeman et al. 2018, Samy et al. 2014).

Currently, there are several user-friendly, high-resolution, global datasets designed for ENM/SDM applications, describing terrestrial (e.g., WorldClim, ENVIREM, Chelsea) and marine (e.g., Bio-Oracle, Global Marine Environmental Datasets) seasonal and annual trends—e.g., mean annual temperature, temperature annual range—and extreme or limiting environmental factors—e.g., temperature of the coldest and warmest month (other sources have been compiled[3]).

The WorldClim database (Fick and Hijmans 2017, Hijmans et al. 2005, Title and Bemmels 2018) is one of the most commonly used sources for global climate data. It offers nineteen climatic data layers (Table 1) representing annual trends, seasonality, and limiting environmental factors. These layers are open source gridded climate data with spatial resolutions ranging from 30 arc-seconds to 10 arc-minutes (~1 to 19 km at the equator) for historical, present, and future climatic scenarios, allowing the transfer of models to past, present, and future. For a complete explanation of bioclimatic data from WorldClim visit its page[4].

Table 1. Bioclimatic variables available at the WorldClim database.

Code	Bioclimatic variables
BIO1	Annual Mean Temperature
BIO2	Mean Diurnal Range (Mean of monthly (max temp - min temp))
BIO3	Isothermality (BIO2/BIO7) (×100)
BIO4	Temperature Seasonality (standard deviation ×100)
BIO5	Maximum Temperature of Warmest Month
BIO6	Minimum Temperature of Coldest Month
BIO7	Temperature Annual Range (BIO5-BIO6)
BIO8	Mean Temperature of Wettest Quarter
BIO9	Mean Temperature of Driest Quarter
BIO10	Mean Temperature of Warmest Quarter
BIO11	Mean Temperature of Coldest Quarter
BIO12	Annual Precipitation
BIO13	Precipitation of Wettest Month
BIO14	Precipitation of Driest Month
BIO15	Precipitation Seasonality (Coefficient of Variation)
BIO16	Precipitation of Wettest Quarter
BIO17	Precipitation of Driest Quarter
BIO18	Precipitation of Warmest Quarter
BIO19	Precipitation of Coldest Quarter

[3] https://freegisdata.rtwilson.com/.

[4] http://www.worldclim.org.

It is worth mentioning that, despite its high relevance, previous exploratory studies (Campbell 2015, Escobar et al. 2014) detected a number of apparent artifacts in some of the climate datasets: mean temperature of the wettest quarter, mean temperature of driest quarter, precipitation of the warmest quarter, and precipitation of the coldest quarter (BIO 8–9, BIO 18–19, respectively). Thus, these variables are commonly excluded from the pool of selected variables to build a model, but regional checking could be performed before discarding them, if there is interest in using such variables. Moreover, an important point to be made is that, regardless of the number and type of climatic variables to be used, all layers must have the same spatial extent, resolution, size (number of pixels, or number of rows and columns), coordinate reference system, and be saved in "ascii" format, which results after masking variables to the calibration area (Section 1.2.1). If other modeling algorithms are used the format of the variables could vary; for instance, if the layers are going to be loaded to R, almost all formats are supported thanks to packages like *raster*, and *rgdal.*

1.3.1. Masking variables

After creating the **M** shapefile that will be used in the study, we mask the variables to create the environmental layers that represent our calibration area. Here, we describe the steps to perform this task in QGIS:

1. In the QGIS Toolbox, under "Raster Toolbox", use the tool "r.mask.vec". Select the shapefile created for the calibration area ("d_auratus_M" in our case), and the variables to be masked. Select a new folder where the calibration layers will be saved (Fig. 6). It is important to keep in mind that the output must have the same name than the variables that will be used as projection layers (e.g., if you are masking the layer "BIO1", the output name will be "BIO1").

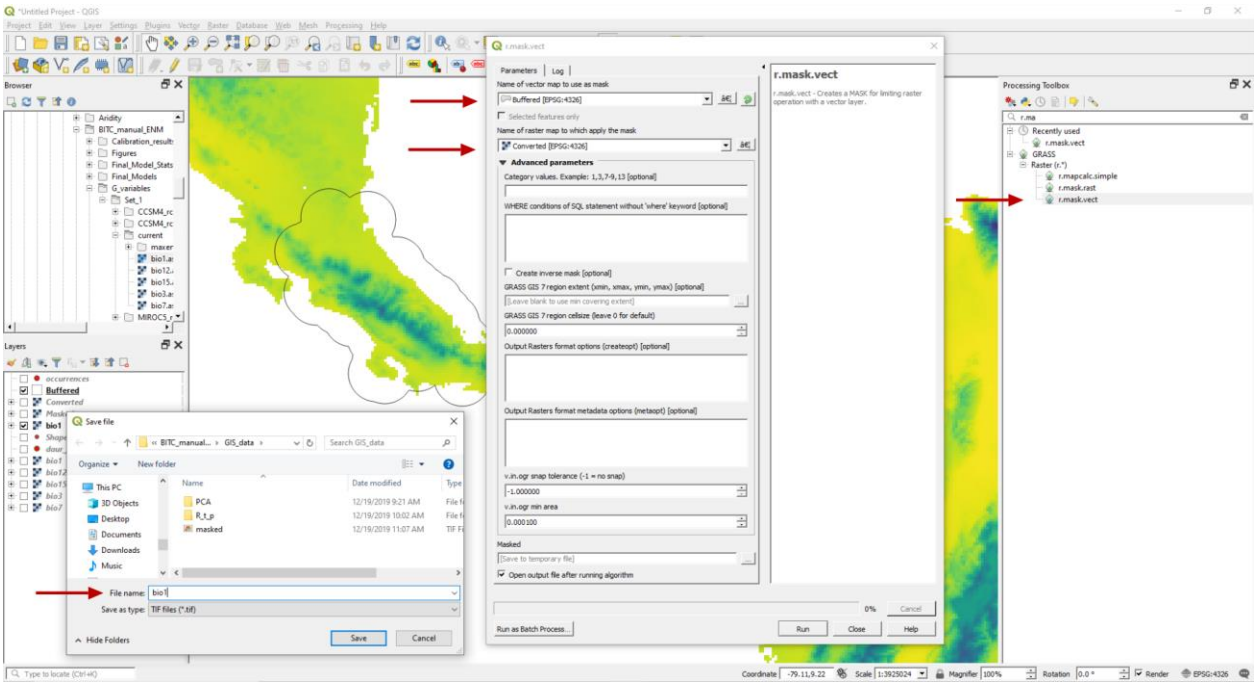


Figure 6. QGIS window showing the use of the Toolbox to mask the variables to the area of calibration.

2. The masked layer will be in "GeoTif" format, which must be transformed to "ascii" format to be able to use it in Maxent. For that, in the Raster conversion Toolbox select "Translate". Then, select the raster layer (in "GeoTif" format) previously created as input, and, under

"Converted", select a name for your layer and the folder where the output file should be stored (Fig. 7).

Both processes, masking and converting formats, can be done in batch, to make the process faster. Moreover, this routine can be performed in R (see Supporting information S2).

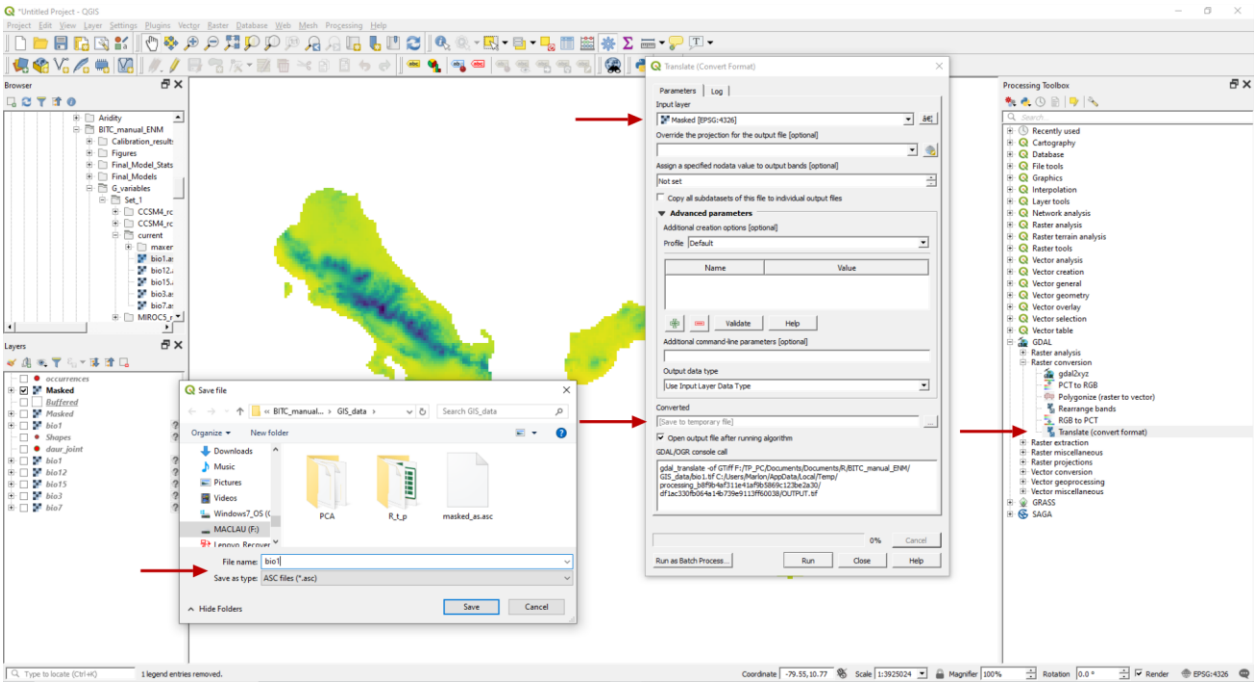


Figure 7. QGIS window showing how to use the Toolbox to convert a raster layer from "GeoTif" to ascii format.

1.3.2. Variable selection

The selection of environmental variables is a crucial step when building an ENM/SDM. Whether large or small subsets of environmental variables are used, they can greatly affect model predictions (Braunisch et al. 2013, Synes and Osborne 2011, Title and Bemmels 2018). For example, models built with highly collinear variables (i.e., variables with linear dependence) are at an increased risk of overfitting and over parameterization (Dormann et al. 2012). In this case, models may behave unexpectedly when projected to other periods or geographic regions because they may encounter combinations of variables that have no analog in the model training (Dormann et al. 2012, Owens et al. 2013, Warren et al. 2014). The addition of environmental dimensions could decrease the chances to find analogous conditions in other scenarios. Moreover, natural-history-driven understanding is crucial in developing statistical models of ecological niches (Rödder et al. 2009), the life history of the species have direct relevance to ecological or physiological processes determining species distributions. Therefore, ideally, one would consider biologically meaningful variables.

Variable reduction approaches can minimize model overfitting and improve model transferability (Jimenez-Valverde et al. 2009, Warren et al. 2014). In general, the number of relevant variables may be reduced either through a statistical analysis, by selecting variables from ecological theory that are likely to be important given the physiology and life history of the organism in question (Rödder et al. 2009, Synes and Osborne 2011), or by using a combination of these criteria. In the following subsections, we explore some commonly applied approaches to select the best combination of variables to build ecological models. Other approaches and its implications in model performance can be consulted in Cobos et al. (2019c).

Based on the techniques presented in this section (correlation analysis and jackknife analysis), as well as on the knowledge on physiological and ecological needs of *D. auratus*, we selected five variables to build our models: annual mean temperature (BIO1), isothermality (BIO3), temperature annual range (BIO7), annual precipitation (BIO12), and precipitation seasonality (BIO15). Two other sets of variables were created by excluding one distinct variable at a time (set 2 excludes BIO5, set 3 excludes BIO3), to be used during model calibration. We produced these other two sets of variables to use them during the model calibration exercise described in the next section, as this has been suggested as an appropriate way to select environmental predictors based on quantitative metrics of model performance (Cobos et al. 2019c).

1.3.2a Variance inflation factor

Multicollinearity in multivariate regressions is the phenomenon in which one predictor variable in the model can linearly predict another variable used as predictor in the same model. This problem implies redundancy in the model and consequently can lead to bias in the predicted outcome. Although the general prediction is relatively safe, interpretations regarding the importance of each variable to the model or effects of independent variables on the model may not be reliable (Farrar and Glauber 1967). In the case of Maxent, Feng et al. (2019) concluded that this SDM is robust to multicollinearity among predictor variables in model calibration, however, multicollinearity can negatively affect model transferability.

The variance inflation factor (VIF) is a measure of the amount of multicollinearity in a set of predictor variables. VIF values range from one to infinity and denote how much the variance regression coefficient is inflated by the existence of correlation among the predictor variables in the model. A VIF of 1 implies no dependence among the environmental variables, while a rule of thumb is that a VIF exceeding 10 indicates high multicollinearity (Chatterjee and Hadi 2000). Other values of VIF are sometimes used as cutoffs (Sheather 2009).

You can calculate the VIF using linear regression in a stepwise approach, where the calculation is done using all the predictor variables and eliminating all the variables that produce VIFs greater than 10. Then, run the calculations again with the new reduced list of predictors; the process continues until you end up with a list of predictor variables all with VIFs smaller than 10 (e.g., Guisan et al. 2017, Jara et al. 2019). The analysis can be implemented in R through the package "usdm" and the function "vif", over a "RasterStack" object (a complete guide to the function is available[5]; see also Cobos et al. 2019c for the use of VIF with Maxent-like general linear models).

1.3.2b Correlation analysis

Another way of assessing multicollinearity is by examining correlations among the predictor variables. With the purpose of eliminating or reducing the amount of redundant information among predictor variables, we recommend analyzing the Pearson correlation coefficients among the environmental variables and evaluating the strength of their relationship within the space where the model is being calibrated. This analysis can be performed in QGIS or in R. In QGIS—with SAGA integrated—the process consists in getting the values of all pixels of each variable (masked to **M**; using the tool "Raster values to points", Fig. 8) as a table and then using these values (or a significant sample of them) to perform a correlation analysis in a statistical software (see Supporting information S2 for a routine in R). Given that there is a lack of an overall agreement in the field on how to choose a lower bound or threshold of admissible correlation (e.g., Nuñez-Penichet et al. 2019, Ramírez-Gila et al. 2018), a threshold should be defined by the researcher based on the central question of the study, the amount of data, and the size and location of the area of study. Based on the results of the correlation analysis along with a jackknife approach

[5] https://cran.r-project.org/web/packages/usdm/usdm.pdf.

(see following section), we can select the least correlated variables that contribute the most to the ecological model.

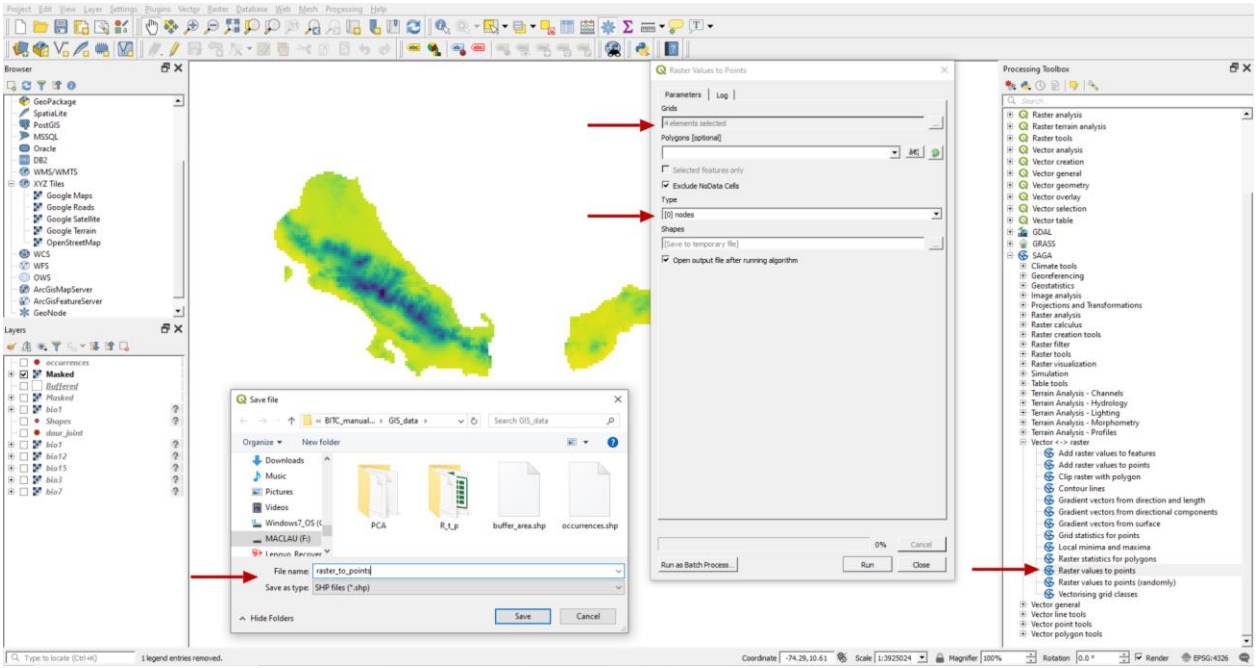


Figure 8. QGIS window showing the tool used to obtain values of raster layers in points, which table will be used to perform the correlation analysis.

1.3.2c Jackknife in Maxent

In statistics, jackknife is a resampling technique useful for variance and bias estimation. In order to calculate the jackknife estimator of a parameter, estimates of the parameter are calculated by systematically leaving out one observation from the dataset, and then, the aggregation of these calculations leads to the jackknife estimator (Efron and Stein 1981).

Getting estimates of which variables are most important in the model is possible through a jackknife analysis in Maxent by selecting the option “Do jackknife to measure the variable importance” on the initial window of the Maxent GUI interface (Fig. 9). With that, the software systematically excludes a variable in each run and the model is created with the remaining variables. Then, a model is created using each variable in isolation, and another one by using all variables. According to Phillips et al. (2006), each step of Maxent increases the gain of the model by modifying the coefficient for a single feature, assigning the increase in the gain to the environmental variable(s) that the feature depends on. After the running, jackknife results are saved in a “csv” file called “maxentResults”. This table has several columns but the ones that are related to the jackknife analyses are “Regularized training gain”, “Training gain without X”, and “Training gain with only X” (where X is the variable of interest, e.g., BIO1). The first of these columns contains the gain when the model is fitted with all the variables, and the other two columns contain the values of contribution when the model is fit without that variable and with only that variable, respectively. These results are also displayed in a graph and when using the Maxent GUI interface, a figure is created so that the comparison of light blue and dark blue bars allows identifying the effect of each variable on the regularized gain of the model.

It is important to highlight that the jackknife test shows which variables have the most useful information independent of the others, not considering the biological role of variables to the focus species, as the test simply shows the best variables for the model, but not necessarily the relevant variables for the species being modeled. The jackknife results from Maxent will help users

to identify what happens to model performance with all variables, excluding each variable, and using each variable independently (e.g., Moreno et al. 2011, Nuñez-Penichet et al. 2016, Raghavan et al. 2019). These three results are important to decide which variables to include, and results from correlation analyses can also help to select variables not only with high contributions but also not correlated.

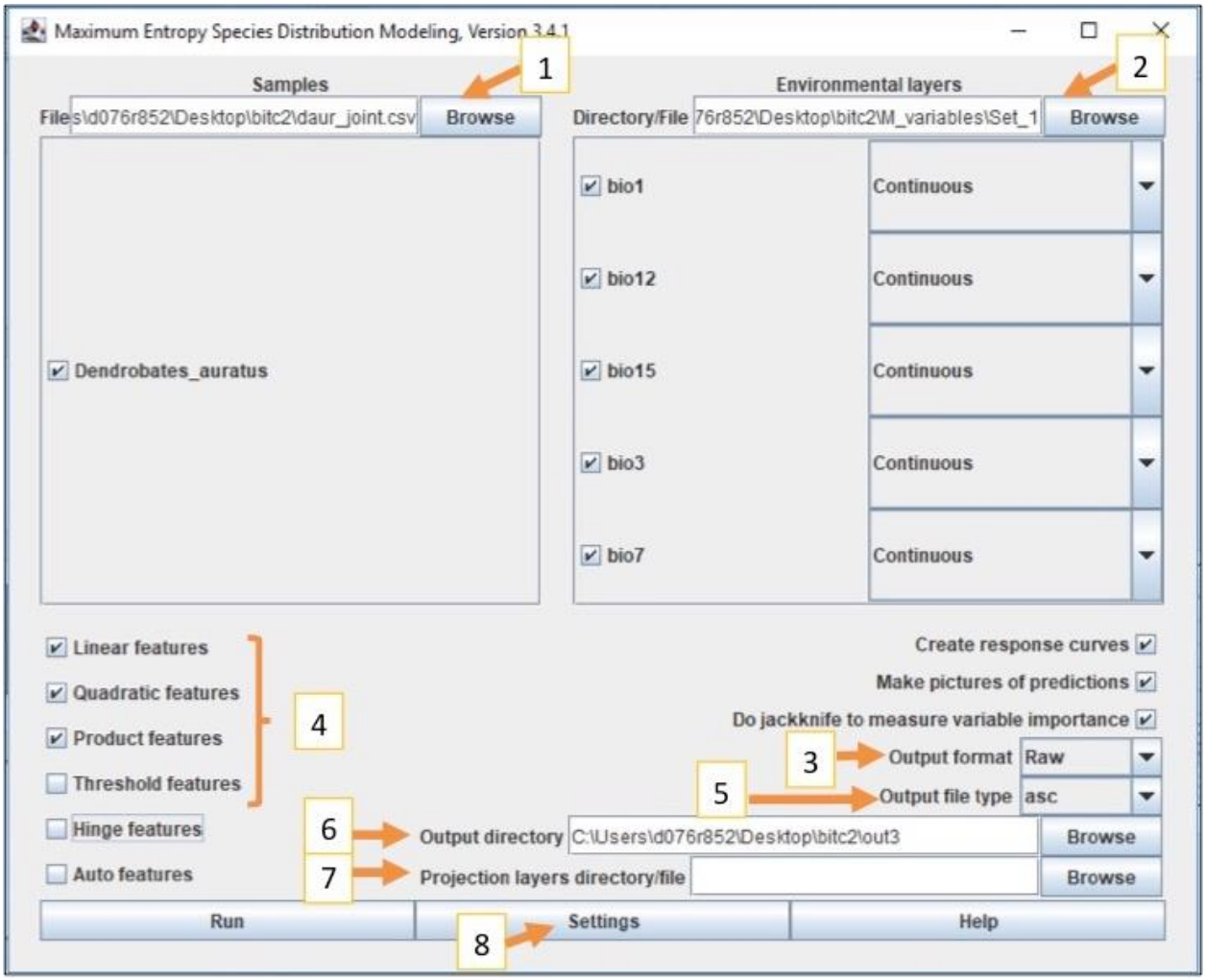


Figure 9. Initial screen of Maxent GUI interface. Numbered boxes indicate settings that are modified to create ecological models of suitability.

1.3.2d Principal Component Analysis

Principal component analysis (PCA) is a statistical procedure that uses an orthogonal transformation to map a set of observations of possibly correlated variables into a set of values of linearly uncorrelated variables called principal components (Pearson 1901). In ENM/SDM, applying this method is a common practice when it comes to reducing dimensionality and collinearity among environmental layers (e.g., Cobos and Alonso Bosh 2018, Freeman et al. 2018, Ingenloff et al. 2017, Simões et al. 2017).

The disadvantage of this method is that you will lose the identity of variables as you will use the principal components instead. The analysis can be performed in QGIS or other GIS software, as well as in R. In QGIS—with GRASS—this analysis can be done using the toll "i.pca", which generates raster layers representing the principal components obtained (see Supporting information S2 for a routine in R). If you are going to use projection layers for a transferability experiment, the principal component loadings obtained for the present layers are then used to transform the future-climate variables (see R routine for an example). Some R packages to perform PCAs that allow using the loadings to transform layers from other scenarios are listed in

Supporting information S1 (e.g., “kuenm”, “ntbox”, “ENMGadgets”; Cobos et al. 2019b, Barve and Barve 2014, Osorio-Olvera et al. 2020).

2. Building an ENM in Maxent

Following the preparation of the occurrence and environmental datasets, we can start with the ENM. In this section, we provide a short description of best practices to create the ENM and describe the calibration, evaluation, and processing of models. The steps in this section were performed in Maxent GUI and the R package “kuenm”. In the following sections, we will refer to steps such as selection of regularization multiplier, feature classes (i.e., model parameters) and evaluation (e.g., PartialROC analysis). The R codes we used to perform this part of the analyses can be found at the Supporting information S2.

2.1 Using Maxent GUI

The user Maxent GUI interface is freely available online[6]. The interface allows the exploration of diverse settings available that can (and should) be modified accordingly to the question of the study at hand. Moreover, as one of the most common algorithms in the field, many of its settings have been largely explored and some level of agreement exists in how to develop a reliable, and most importantly, replicable, model (Araujo et al. 2019, Feng et al. 2019). For instance, Maxent allows four distinct output formats: raw, logistic, cumulative, and cloglog (Elith et al. 2011, Merrow et al. 2013, Phillips et al. 2017). The different types outputs are transformations from the raw probabilities, including a logistic transformation of the raw values of suitable environmental conditions (Peterson et al. 2011). Raw outputs represent relative abundance, which is the ratio of presence sites and the marginal density of covariates across the study area, as an estimate of the relative suitability of one place *vs.* another (Elith et al. 2011). The logistic output provides an index of suitability which should be interpreted hierarchically also considering whether calibration and model evaluation occur in a study region where biotic factors and the calibration area do not affect the species’ distribution (Peterson et al. 2011). The cumulative output is defined in terms of omission rates (Phillips et al. 2006), the cumulative value of the Maxent output represents the percentage of occurrences below that threshold.

Thus, if the test data are drawn from the Maxent distribution, the predicted omission rate is the percentage of the cumulative value divided by 100. However, under conditions of strong sample selection bias, we can expect significant deviations from predicted omission rates (Phillips 2008). The cloglog output is a transformation of the raw output to produce an estimate of occurrence probability (Phillips et al. 2017). This output is an implementation derived from an interpretation of Maxent as a non-homogeneous Poisson process (Aarts et al. 2012, Fithian and Hastie 2013). Further information on Maxent settings can be found on Phillips et al. (2006), Elith et al. (2011), Merow et al. (2013), and Phillips et al. (2017).

Following is a step-by-step guide to start developing models with Maxent GUI (Fig. 9):

1. In the initial screen of Maxent GUI interface, add the occurrence data processed in previous steps (Section 1.1) under the field “Samples” (field 1). Remember that this is usually a “csv” file with three columns: “species”, “longitude”, and “latitude” with the prepared occurrence records.
2. Add the environmental predictors masked to the studied calibration area (Sections 1.2 and 1.3) under the field “Environmental layers” (field 2).
3. Select the output format as “raw” under the field “Output format” (field 3; Section 2.2).

[6] https://biodiversityinformatics.amnh.org/open_source/maxent/.

4. On the left-hand side, let us check the boxes “Linear features”, “Quadratic features”, and “Product features” (field 4). These parameters allow the use of different transformations to the environmental predictors in order to fit the data (Elith 2011).
5. Under the “Output file type” (field 5) select the format “asc”.
6. In the “Output directory” (field 6) select the folder where the output will be deposited. In case the user wishes to transfer models to different periods or regions, so the projection layers should be indicated under “Projection layers directory/file” (field 7).
7. Next, click on the option “Settings” (field 8). This will take the user to another window where basic, advanced and experimental settings can be chosen (Fig. 10).
8. The option “Random seed” (field 9) will assure a random partition of occurrences for each run, avoiding repetition and bias of the data selected while creating models in each run.
9. Under the field “Random test percentage” (field 10) type in “50”, which corresponds to the percentage of data used to the internal training/testing split to build models (See next section).
10. Under the field “Regularization multiplier” (field 11), you can select any value; here we are using a multiplier of two. This option allows the user to select the value of the regularization multiplier, which limits the complexity of the model and generates a less localized prediction (Phillips and Dudík 2008).
11. Select the number of replicates desired under “Replicates”. In this case one replicate since we are creating our first model.
12. Under the field “Replicate run type” (field 12) select “Crossvalidate”.
13. Close the settings window (field 13) and click “Run” on the bottom left of the Maxent main window (Fig. 10).

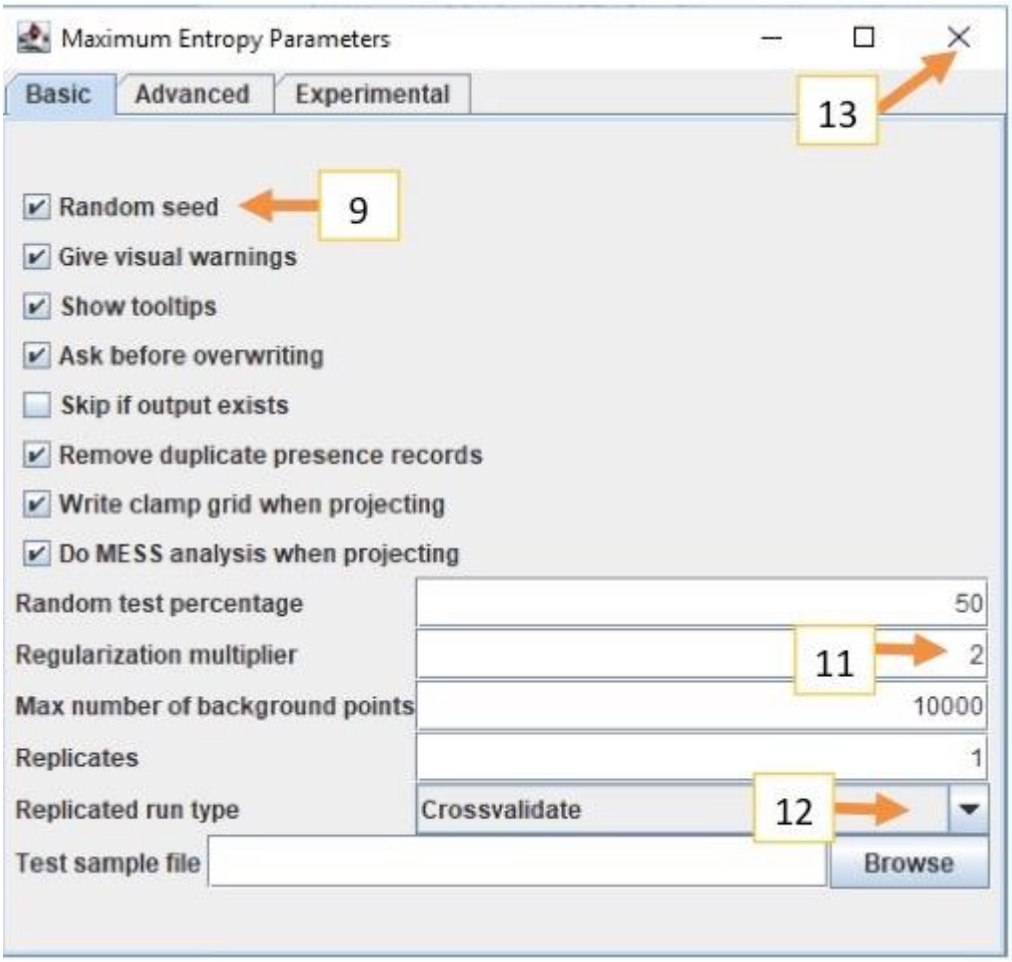


Figure 10. Settings window that should be modified from its default counterparts.

If you explore the output folder, you will see different elements including an "HTML" file with the name of the species ("Dendrobates_auratus" in our example). By opening that file you will be able to review the settings of the model (at the end of the "HTML" file) and different plots developed by the software including the raw output of suitability, the response curves of each variable, the jackknife analysis of the variables (Section 1.3.2c), among others.

Now, as you see up to this point, we can alter different settings from the GUI Maxent interface; some of them will affect heavily on the output models, specially the selection of "Feature classes" and the "Regularization multiplier". In order to determine the best combination of parameters for a reliable model, we should repeat this process multiple times changing the combination of parameters for each time (e.g., Regularization multiplier = 1 vs. 2 vs. 3, etc.). Then, we have to establish some criteria to determine which of these models perform the best, given the provided data. We called this step model calibration and we will discuss it in the next section. Although this process can be done manually, the time needed for this task is usually long and, in some sense, error-prone and difficult, given the large number of models needed.

2.2 Model calibration

Model calibration is a process in which a large number of candidate models are tested to identify the parameter settings that lead to the best fitting models to the question and data at hand. In Maxent, candidate models can be created through multiple runs using combinations of distinct values of regularization multipliers, feature classes, and preferably, more than one set of environmental variables (see Cobos et al. 2019b, Muscarella et al. 2014, Peterson et al. 2018, Warren et al. 2010). Regularization multipliers help smoothing the model, which avoids fitting suitability only to climatic values that are very similar to the ones in the occurrences. Higher values of regularization multipliers make the model more permissive in the sense that, to be considered as suitable, multivariate climatic values could be less similar to the ones that strictly correspond to the occurrences. Feature classes correspond to the response type of suitability values to each variable. Common response types are linear, quadratic, and product; other response types in Maxent include threshold and hinge (for further details on regularization multipliers and response types see Elith et al. 2011 and Merow et al. 2013).

Model selection should be based on metrics of model performance that are adjusted to the algorithm used (Lobo et al. 2008, Velasco and González-Salazar 2019). We recommend that models should be selected using at least three criteria in the following order: statistical significance (partial ROC; Peterson et al. 2008), predictive ability (omission rates; Anderson et al. 2003), and model complexity (AICc; Warren and Seifert 2011). The metric AICc is used here only after checking for statistical significance and prediction ability as it has been found to perform poorly as a test of geographic prediction accuracy (Velasco and González-Salazar 2019). However, this metric (AICc) is a good indicator of model complexity as it accounts for the number of parameters in models when measuring how well the predicted values fit the data. Thus, models selected by these three metrics will maximize model performance and simplicity (Cobos et al. 2019b). To calculate the two first metrics the whole set of occurrences (previously cleaned) need to be split into training and testing data; a model is created with the training data and then significance and omission rates are calculated using the data saved for testing. The third metric (AICc) is calculated using the whole data set (Cobos et al. 2019b, Muscarella et al. 2014).

For our study species, *D. auratus*, we created 120 candidate models by combining three sets of environmental predictors (see variable selection), eight values of regularization multipliers (0.1, 0.4, 0.7, 1, 2, 3, 4, 5, 6, 7), and five combinations of three feature classes (linear = l, quadratic = q, product = p; lq, lp, q, qp, lqp). We evaluated candidate model performance based on significance (partial ROC; with 500 iterations and 50 percent of data for bootstrapping), omission rates (E = 5%; Peterson et al. 2008), and model complexity (AICc). Best models were selected

according to the following criteria: (1) significant models with omission rates ≤ 5%, and (2) models with delta AICc values of ≤ 2.

2.2.1 Practice on model calibration

As mentioned earlier, an exercise of manual model calibration is possible but usually prohibitively long. Here we performed the model calibration process using the R package "kuenm" which also allows testing different environmental variables associated with our studied system (Supporting information S1, Cobos et al. 2019b):

1. Install the kuenm R package following instructions[7].
2. Prepare "csv" files containing the occurrence data:
   (a) One file containing all occurrence records that have been filtered and thinned.
   (b) A second file containing the subset of occurrences for training the models.
   (c) A final file containing the subset of occurrences for testing models.

   Distinct ways to generate subsets of training and testing datasets exist (Muscarella et al. 2014). All the "csv" files need to have species name, longitude, and latitude, as columns, in that specific order, as mentioned in Section 1.1 and Fig. 2. You can split the occurrences manually, or you can use the function "kuenm_occsplit" available in the R package "kuenm" (Supporting information S2).
3. Prepare variables for model calibration. After excluding predictors with correlation levels above the desired limit and after deciding the potential biological relevance of these variables, a reduced number of variables will remain (Sections 1.3). The selected variables can be split in two sets to test the effects of adding or excluding them. The number of sets depends on the goal of the study. For example, in this case, three sets can be tested. A folder containing these three sets of variables in three sub-folders named as the variables in each set need to be created. Copy the corresponding variables inside each sub-folder (an example of folders organization for applying "kuenm" functions).
4. Run candidate models. Now all data is ready for the calibration process. Start a session in R and load the "kuenm" R package. Prepare arguments as in the example of the "kuenm_cal*"* function (Supporting information S2); make appropriate changes for coincidence of directories and names (reading the function's help is recommended to understand how each argument needs to be defined). Run the function and check that the models are being created in the corresponding folders of the working directory. Notice that the software Maxent needs to be stored in a directory of your computer to be able to produce the models.
5. Run evaluation of candidate models using the function "kuenm_cfeval*"* from "kuenm". The evaluation of candidate models can be performed together with the creation of candidate models, if the argument "wait" is set to FALSE in the "kuenm_cal" function (Supporting information S2). In our exercise, 24 candidate models were built, but only 21 were statistically significant (those are shown in Table 2). Model selection is performed immediately after model evaluation with the same function.
6. A new folder containing the results from calibration will be created. This folder will contain: the general statistics of the process, the results of evaluation for all candidate models, a plot of the evaluation results, a file containing information for the selected parameter settings, and an "HTML" file that helps with interpretation. In our exercise with *D. auratus*, all 120 models were significant, 16 of those met the omission rate criteria, and among those significant low omission models, only one of them met the criteria for being selected as the best model (M_2_F_lp_Set_1). The values of evaluation metrics for the selected model were: <0.05 for *P* value, 0.043 for omission rate, and 3217.72 for AICc.

[7] https://github.com/marlonecobos/kuenm.

Table 2. Results of candidate model evaluation filtered according to model selection criteria. Only results for models that met the criteria of significance and omission are presented.

Models	Mean AUC	Partial ROC (*p*-value)	Omission rate (E = 5%)	AICc	Delta AICc	W AICc	No. Parameters
M_2_F_lp_Set_1	1.114	0.000	0.043	3217.718	0.000	9.3E-01	8
M_0.4_F_lq_Set_1	1.200	0.000	0.043	3223.026	5.308	6.6E-02	10
M_0.4_F_qp_Set_2	1.283	0.000	0.043	3303.836	86.118	1.9E-19	8
M_0.1_F_lp_Set_2	1.278	0.000	0.043	3315.601	97.883	5.2E-22	8
M_1_F_lq_Set_3	1.163	0.000	0.043	3323.065	105.347	1.2E-23	6
M_0.4_F_lp_Set_3	1.193	0.000	0.043	3327.254	109.536	1.5E-24	8
M_0.1_F_lp_Set_3	1.196	0.000	0.032	3327.856	110.138	1.1E-24	9
M_4_F_lqp_Set_3	1.157	0.000	0.043	3348.107	130.389	4.5E-29	6
M_2_F_lp_Set_3	1.144	0.000	0.043	3351.269	133.551	9.3E-30	7
M_3_F_lp_Set_3	1.132	0.000	0.043	3364.314	146.596	1.4E-32	7
M_3_F_lq_Set_3	1.171	0.000	0.043	3373.667	155.949	1.3E-34	5
M_4_F_lq_Set_3	1.153	0.000	0.043	3387.801	170.083	1.1E-37	6
M_5_F_lq_Set_3	1.142	0.000	0.043	3396.295	178.578	1.6E-39	5
M_3_F_q_Set_3	1.128	0.000	0.043	3433.152	215.434	1.5E-47	4
M_4_F_q_Set_3	1.129	0.000	0.043	3434.499	216.781	7.9E-48	4
M_5_F_q_Set_3	1.137	0.000	0.043	3436.021	218.303	3.7E-48	4

2.3 Final model in Maxent

After selecting the set of parameters that produce the best models, a final model can be created to represent the niche of the species. Sometimes more than one set of parameter settings result from the calibration process, in those cases, using all of them is recommended unless another criterion of selection exists (e.g., models with low omission rates but with less area predicted as suitable). These models should be created with replicates (i.e., multiple repetitions) to add variability to predictions, and the type of replicates must be selected depending on the final goal of the research. In general, bootstrap replicates are recommended (Cobos et al. 2019a, Kass et al. 2018, Muscarella et al. 2014).

From the model calibration exercise, one model was selected as the best. This model can be created using Maxent GUI following the steps described before or they can be done following the workflow in "kuenm". To create the selected model with replicates and projections use the function "kuenm_mod*"* in the "kuenm" package (Supporting information S2). In order to run this function, we have to specify extra arguments including the output format (i.e., logistic), the replication type (i.e., bootstrap), the number of replicates (e.g., 10), and whether or not we are going to project our final model to different areas (More details can be found through the function's help in R). The function will automatically retrieve the information from the folder "Calibration results" that was created in your working directory during the calibration steps (Section 2.1.1).

After executing the "kuenm_mod" function, a folder with the name "Final Models" will be generated with the results. For the purposes of this manual, we are focusing on the model "M_2_F_lp_Set_1", considering that this model was selected during model calibration. However, the function will create folders with results for all the models with parameters that achieve a delta AICc less than two if multiple models are selected as good candidates (Section 2.1; Cobos et al. 2019b). In each folder, you will find the bootstrap replicates, together with raster files representing summary statistics (e.g., average, maximum, minimum, etc.) for the specific model.

2.4 Processing Maxent outputs

The outputs of the final models can be explored using the "HTML" file produced by Maxent in which the suitability patterns can be explored in geography as well as for each of the predictors. In the selected output folder (Fig. 9, field 6), raster files in "ascii" format corresponding to the average, standard deviation, median, maximum, and minimum values of the bootstrapped replicates are available. To represent the geographic predictions of final models, it is recommended to use the median of all the bootstrapped replicates (Fig. 11); the average of replicates may not be adequate because this statistic is more sensitive to extreme values, which may be common among replicates (Cobos et al. 2019a). It is also adequate to represent variability in model predictions and the range (i.e., difference of maximum and minimum predictions) is, for instance, a practical way to show how variable the replicates were for the calibration area and the projection scenarios (Fig. 11; for a more detailed reference on how to represent model variability, see Cobos et al. 2019a).

Exploring the response curves for each predictor will help understanding which values of the variables are more or less suitable for the species according to the model. Although model projections are not treated in this manual with detail because our goal is to represent a rather simple modeling exercise, we have projected our model to the neotropics for demonstration of modeling output processing (Fig. 11). Projections involve a series of further analyses and interpretations that users should consider; for instance, selection and explorations of types of extrapolation (see "Extrapolation analysis" and "Model interpretation" sections).

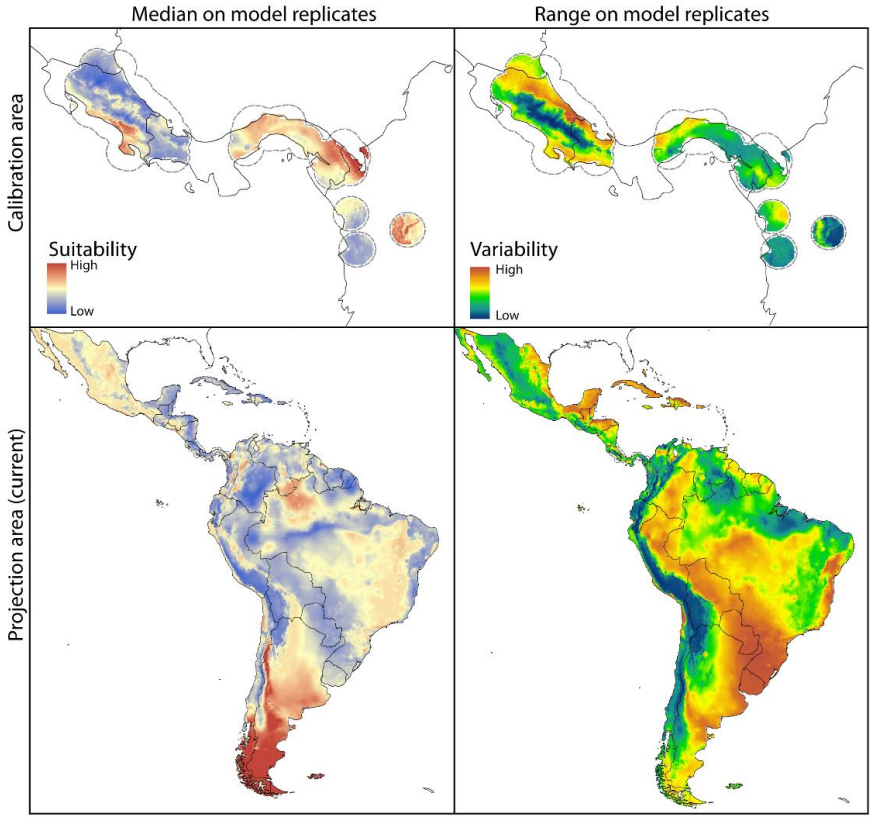


Figure 11. Median and range of replicates for the calibration area and the current projection. Projection was performed with extrapolation and clamping for demonstration purposes only.

2.4.1 Practice on Maxent output processing (Thresholding)

The continuous suitability index derived from the logistic output selected to develop the final Maxent outputs show pixel values ranging from 0 to 1 (Merow et al. 2013; Fig. 11). However, ENM studies usually are dealing with a binary suitable/non-suitable question, which grants the

further processing of the models into binary raster files. To accomplish this step, we can use different rules to binarize Maxent outputs, among them, using the values from the presence points with the less suitable values is a straightforward option. This binarization approach has been called the "lowest presence threshold" or "minimum training presence" rule (Pearson et al. 2007). Among other rules, we can assume a percentage of error in the occurrences used (e.g., 5%, 10%) and use the first value above the number of occurrences expected to be erroneous (those with the lowest suitability values) as the threshold rule (Peterson et al. 2008). The threshold value is an important decision to make as this could change dramatically the extension of areas considered suitable. This is why, we encourage to find an adequate threshold based on a level of error assumed to persist in your data after cleaning occurrences and use that value consistently across the modeling process (i.e., during model calibration and for creating model binary predictions; Peterson et al. 2008).

Here we go through a few steps, performed in QGIS, to set the threshold to obtain the binary prediction of suitable habitats for *D. auratus* in the exemplified transference model to the Neotropics:

1. Load the occurrence data and median output of Maxent for *D. auratus* in your calibration area (see Section 2.1.4).
2. We will export the occurrence points of *D. auratus* to a shapefile format, to transform it into a geospatial vector that can be used in further analyses in QGIS (Fig. 12).

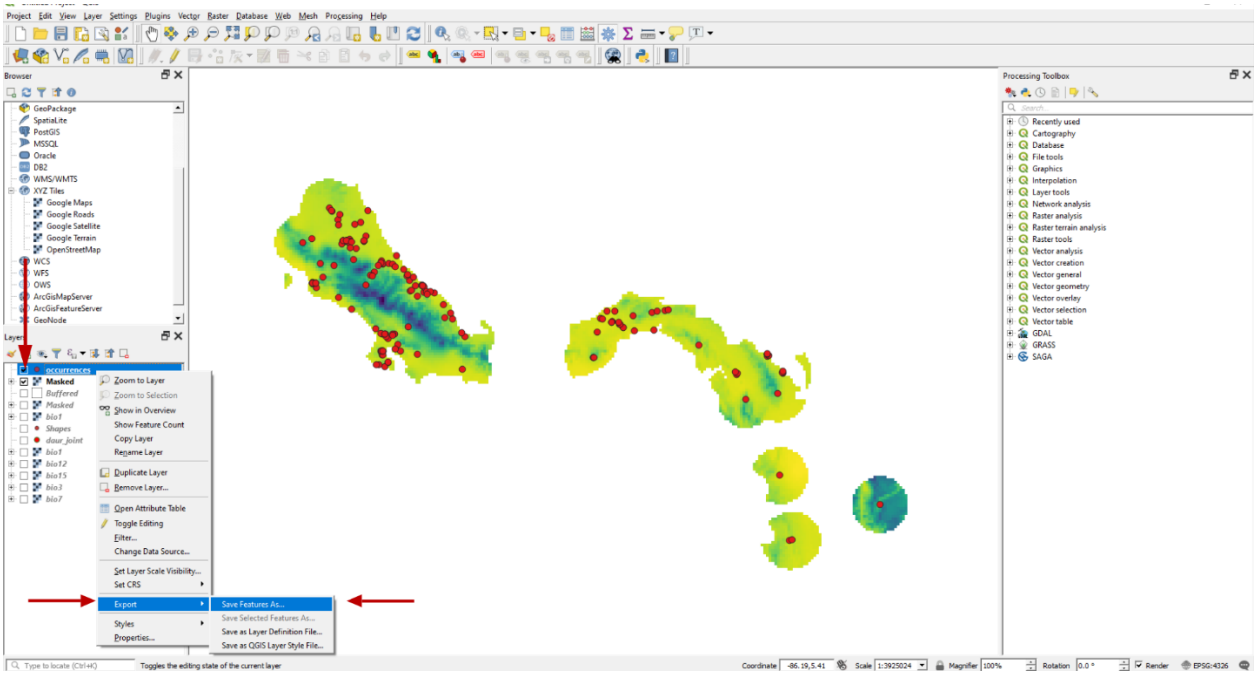


Figure 12. QGIS window showing the process of exporting occurrence data points to shapefile.

3. Now we will extract the values of each occurrence point from the median raster, using the tool "Add values to points", under "vector < – > raster", available in the SAGA Toolbox (Fig. 13). Load the *D. auratus* shapefile and the raster that will have values extracted from the median raster (i.e., D_auratus_med.asc).
4. Set the threshold value used to transform the model into binary. For example, here we will apply the 5% threshold: we chose to leave out the 5% lowest values of our occurrence points (suitability < 0.136; Fig. 14), and reclassify the median raster of the model projected to the Neotropic using the tool "Reclassify by table", under "Raster analysis".

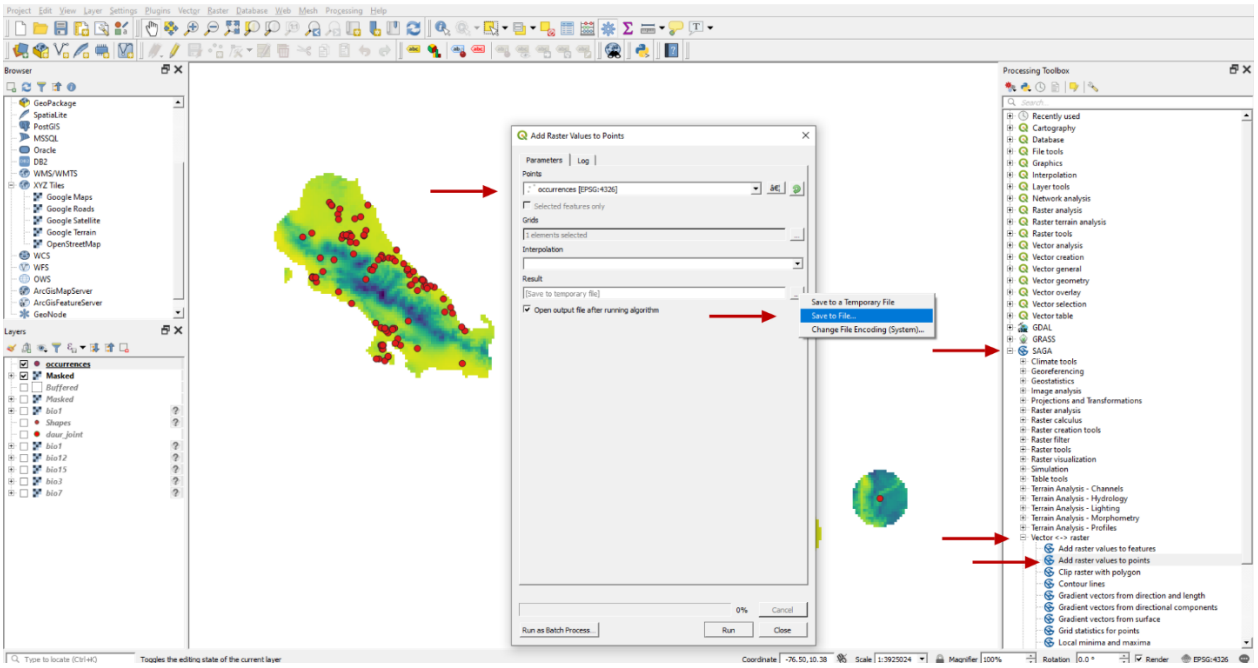


Figure 13. QGIS window indicating the path (red arrows) to followed to find the Add values to points tool in the Toolbox.

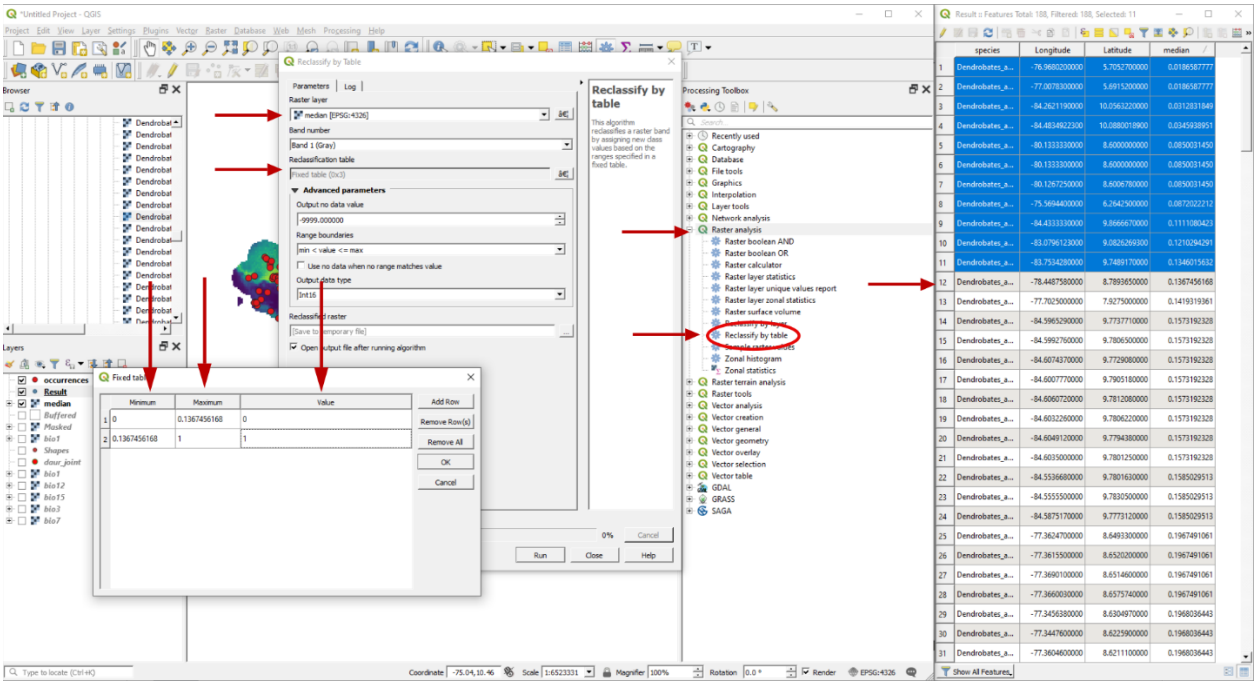


Figure 14. QGIS window showing the process of raster reclassification. Red arrows on the Reclassify window indicate the range of values considered to transform the raster into binary. On the middle panel, the red circle and arrows indicate the location of the reclassify tool. On the right, the table shows the values obtained for each occurrence point and the red arrow indicates the percentage used as threshold. The points highlighted in blue are the 5% with the lowest values.

5. Finally, we click "OK" in the Reclassify window and a binary map will appear (Fig. 15), where a value of 0 indicates unsuitable areas, and a value of 1 indicates suitable areas.

This process can also be performed in R, see Supporting information S2 for R routine.

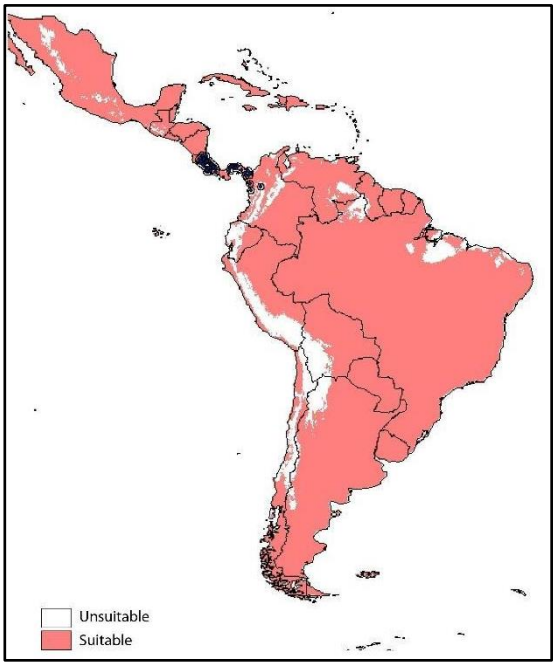


Figure 15. Binary map showing habitat suitability for *Dendrobates auratus* within the Neotropical region.

Extrapolation analysis

The usage of ENM approach ranges through diverse purposes, from estimating the relative suitability of habitat known to be occupied or not known to be occupied by the species, changes in the suitability of habitat over time, and estimates of the species niche (Owens et al. 2013, Peterson et al. 2011, Warren and Seifert 2011). Most of these applications involve the transfer of model predictions to distinct regions and/or periods (Elith and Leathwick 2009, Owens et al. 2013, Pearson et al. 2006, Qiao et al. 2018), which might show values outside the calibration region sampled during training (i.e., strict extrapolation), or portions of environmental space may be within the range of individual variables, but represent new combinations of their values (i.e., combinational extrapolation; Qiao et al. 2018, Owens et al. 2013, Zurell et al. 2012).

Maxent has three distinct options for extrapolation: (1) free extrapolation, (2) extrapolation with clamping, and (3) no extrapolation (Supporting information S3). Under free extrapolation settings, responses in areas environmentally different from the calibration area follow trends in the calibration environmental data. With the extrapolation and clamping setting, the response in areas with environments distinct from those in the calibration area are clamped to levels presented at the periphery of the calibration region in environmental space. Finally, under the no extrapolation setting, the response is set to zero if the environments in transfer areas are extremer than those in areas across which the models were calibrated. Exploring these three distinct types of extrapolation is recommended to understand what the implications of predicting suitability values in projection areas where conditions are non-analogous to the ones in the calibration area would be (Cobos et al. 2019b, Owens et al. 2013).

Strategies to measure extrapolation on model transferences include Multivariate Environmental Similarity Surface (MESS; Elith et al. 2010) and the Mobility-Oriented Parity metric (MOP; Owens et al. 2013). The MESS analysis, implemented in the Maxent software (Phillips et al. 2006), identifies regions of strict extrapolation and provides an index of

environmental similarity between each pixel and the median of the most dissimilar variable in the calibration area. The MOP analysis is a modification and extension of the MESS. It measures distance in environmental space between the different points within the calibration area and the projection area; and, by doing that, considers the variation of environments within the calibration area, hence allowing a better estimation of areas of strict extrapolation (for further information on MOP see Owens et al. 2013). Identifying novel environments and characterizing how ENM algorithms fit response surfaces to conditions extending beyond those over which the model was calibrated, is relevant to provide a view of certainty and uncertainty that should be accorded to different sectors of the projection area (Owens et al. 2013, Zurell et al. 2012).

For our study species model, we estimated areas of strict extrapolation within the projection area using the MOP analysis, which measured the similarity between the closest 5% of the environmental conditions of the calibration area to each environmental condition in the area of transference. To perform MOP analysis, we used the function "kuenm_mop" available in the kuenm R package (see Supporting information S2 for an R script). Areas of projection with lower values of similarity compared to **M** indicate higher extrapolative values and, therefore, higher uncertainty; caution is required when interpreting whether suitability exists or not in such areas (Alkishe et al. 2017). We threshold the MOP result, considering only areas with zero similarity as a strict extrapolation (0 = strict extrapolation, 1 = non-extrapolative areas).

The results indicate that most of Central and South America are areas of extrapolation (Fig. 16). This might be explained by the reduced distributional range of the species, in comparison to the projection area. The discrepancy between size of projection and calibration areas, might precipitate large areas estimated as strict extrapolative, due to lack of environmental variation or diversity of environments within the calibration area. Areas of extrapolation should always be explored and described so that the interpretation of final models can incorporate the levels of uncertainty and extrapolation it might contain, especially in case of a heavily parameterized model.

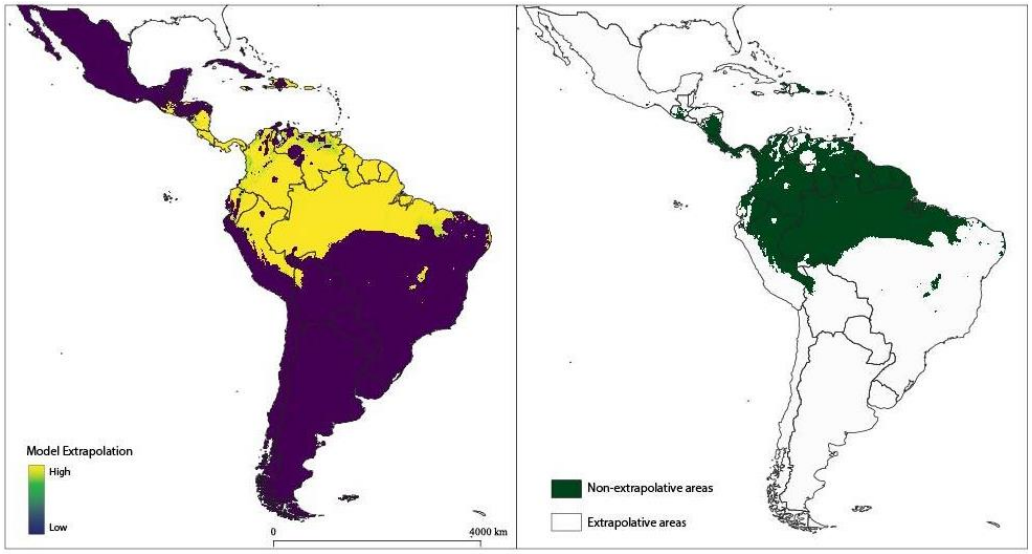


Figure 16. Areas of extrapolation estimated using MOP analysis. Areas of strict extrapolations are shown in the map.

Model Interpretation

As previously mentioned, species distributions are limited by movement capacities, abiotic conditions, and biotic interactions (i.e., fundamental niche; Soberón and Peterson 2005). Their intersection is a complex expression of its ecology and evolutionary history, determined by diverse factors operating with different intensities and scales (Brown 1995, Gaston 2003, Pearson and Dawson 2003). Within this context, the distinction between biotic interactions and abiotic limitations on a species' distribution can be formalized in the distinction between fundamental and realized niches (Hutchinson 1957), where realized niche is whereby a species is excluded from parts of its fundamental niche because of competition and other biotic interactions (Austin et al. 1990, Guisan and Zimmermann 2000).

Correlative ENMs such as Maxent models, focus on relating environmental conditions that meet species' ecological requirements and predict the relative suitability of environments across landscapes of interest to define areas that show conditions which species are likely to maintain viable populations (Araújo and Peterson 2012, Warren and Seifert 2011). It focuses on identifying regions that are similar in terms of the layers provided to those where occurrence points are located (Soberón and Peterson 2005) and, despite observations on species distribution, fundamentally include effects of biotic interactions; these solely represent the realized niche. Thus, ENMs are restricted to approximate the existing fundamental niche, potential geographic distribution of species (Phillips et al. 2006), or suitable areas, given the environment layers used to calibrate the model, and not "probability of occupied area", as mistakenly assumed in some studies (e.g., Fourcade et al. 2014, Syfert et al. 2013).

Conclusions

As ENMs and SDMs have proven useful in a plethora of areas—quantitative ecological studies, evolutionary biology, conservation planning, resource management, public health—understanding and providing tools promoting the appliance of this method is of utmost value to enable analysis, robustness, and replicability. In this manual, we present the step-by-step of the modeling process, seeking to provide a broader understanding and ensure consistency on the generation of ecological and species distribution models.

However, in this manual we share the basics to develop reliable ENM/SDMs; and, in view of the growing literature and daily development of new tools and methods, we highlight the availability of further literature (e.g., Peterson et al 2011 and Guisan et al. 2017), and other freely available sources that serve to complement the information found in this manual. Among them are online courses and journals, which more frequently publish studies on the topic (Supporting information S4).

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Supporting information S1

Table S1. List of alternative tools and software that can be used to perform an ENM/SDM analysis and work with maps and geographic information.

Analysis	Tool/Packages	Software	Source
SDM/ENM analysis	R/kuenm	Open source	www.github.com/marlonecobos/kuenm
	R/ENMeval	Open source	www.cran.r-project.org/web/packages/ENMeval
	R/Wallace	Open source	www.cran.r-project.org/web/packages/wallace
	R/dismo	Open source	www.cran.r-project.org/web/packages/dismo
	R/biomod2	Open source	www.cran.r-project.org/web/packages/biomod2
	R/ntbox	Open source	www.github.com/luismurao/ntbox
	R/ellipsenm	Open source	www.github.com/marlonecobos/ellipsenm
	R/ENMGadgets	Open source	www.github.com/narayanibarve/ENMGadgets
	R/maxnet	Open source	www.cran.r-project.org/web/packages/maxnet
	R/ENMTools	Open source	www.github.com/danlwarren/ENMTools
Geographic Information Systems (GIS)	Quantum GIS	Open source	www.qgis.org
	DIVA-GIS	Open source	www.diva-gis.org
	GVSIG	Open source	www.gvsig.com/en/home
	ArcGIS	Commercial	www.esri.com/software/arcgis

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Supporting information S2

Table S2. Links and list of R scripts included in the manual.

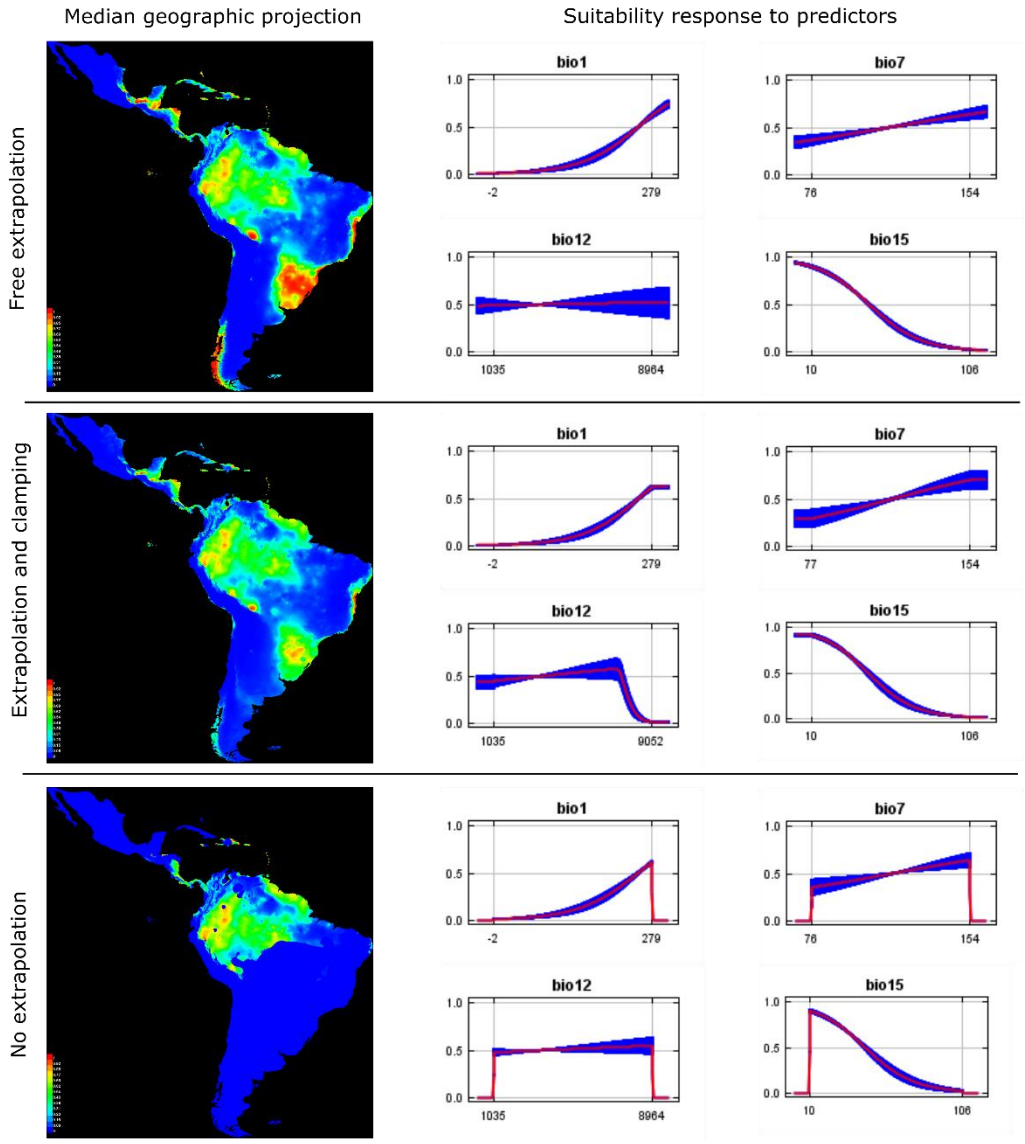
Repository / Process	Link
General repository of scripts	https://github.com/marlonecobos/ENM_manuals
Thinning occurrences and splitting training and testing data	https://github.com/marlonecobos/ENM_manuals/blob/master/Data_cleaning/Thinning_split_train-test.R
Creation of calibration area (M)	https://github.com/marlonecobos/ENM_manuals/blob/master/M_hypotheses/Construction_of_simple_Ms.R
Masking layers to M	https://github.com/marlonecobos/ENM_manuals/blob/master/Variables_processing/Masking_variables_with_M.R
Variable correlation analysis	https://github.com/marlonecobos/ENM_manuals/blob/master/Variables_processing/Variables_correlation_evaluation.R
Principal component analysis	https://github.com/marlonecobos/ENM_manuals/blob/master/Variables_processing/PCA_raster_and_projections.R
Partial ROC, omission rates, and AICc	https://github.com/marlonecobos/ENM_manuals/blob/master/ENM_process/Evaluation_metrics.R
Model calibration	https://github.com/marlonecobos/ENM_manuals/blob/master/ENM_process/Model_calibration.R
Final model creation	https://github.com/marlonecobos/ENM_manuals/blob/master/ENM_process/Final_models.R
MOP analysis	https://github.com/marlonecobos/ENM_manuals/blob/master/Post_modeling/MOP_analysis.R

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Supporting information S3

Figure S1. Examples of geographic projections and response curves obtained for the Neotropics when projecting the model for *Dendrobates auratus* using three distinct types of extrapolation in Maxent.



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Supporting information S4

Table S3. List of freely available sources that serve to complement the information found in this manual. The table shows a list of online resources and journals which more frequently published studies on the topic.

Repository / Process	Link
Ecological Niche Modeling Course 2020 (English)	https://www.researchgate.net/project/ENM-2020-Online-Course-in-Ecological-Niche-Modeling?esc=ProjectShareMail
Ecological Niche Modeling Course 2018 (Spanish)	http://biodiversity-informatics-training.org/bi-curriculum/curso-modelado-de-nicho-ecologico-2018/
Biodiversity Training Curriculum	http://biodiversity-informatics-training.org/bicurriculum/#squelch-taas-tab-content-0-8
Journals	*Ecography*; *Ecology Letters*; *Ecology and Evolution*; *Diversity and Distributions*; *Journal of Biogeography*, *Frontiers in Biogeography*, *Global Ecology and Biogeography*, and *Biodiversity Informatics*, among others